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Renewables but unjust? Critical restoration geography as a framework for addressing global renewable energy injustice

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ABSTRACT

A global transition toward a sustainable energy system, incorporating for example Renewable Energy Technologies (RETs), is essential for decarbonizing electricity production, meeting energy demands, and mitigating the impacts of climate change. However, the growing scale of renewable energy development has exacerbated local environmental and social challenges; improper assessment of RETs has led to recorded conflicts and resource injustice in transitioning communities. The purpose of this study is to analyze global cases of renewable energy development resulting in conflict and environmental injustice, and to propose Critical Restoration Geography (CRG) as a framework strategizing for pre-emptive avoidance of RET-related injustices. Evidence of global environmental injustice in RET development was explored using recorded conflicts from the Global Atlas of Environmental Justice (EJAtlas). We synthesized global variations in affected demographics, land area and conflict resolution with respect to achievement of environmental justice by RET type (wind, solar, biomass, geothermal). Based on analysis of GEJA's 102 recorded cases of RET-related environmental (in)justice from 2001 to 2021, justice was either not achieved or ambiguous in 55 and 20 cases. Drivers for these injustices include displacement of Indigenous communities, exclusion of communities from decision making processes, and protection of business interests over biodiversity and community needs. The proposed CRG framework details seven principles for avoiding environmental injustice in global RET development; including recognition and deconstruction of power dynamics, incorporation of multiple knowledge systems, and promotion of social justice. These principles serve to inform environmentally just approaches to policymaking for future RET development in any geographical context.

1. Introduction

Transitioning to a sustainable energy system, in part by adopting Renewable Energy Technologies (RETs), is essential for decarbonizing electricity production, meeting energy demands and mitigating impacts of climate change [1,2]. RETs such as wind, solar, geothermal, and biomass have been adopted by many public and private organizations worldwide, often motivated by pursuance of 'net-zero' greenhouse gas emissions [3,4]. This motivation has resulted in 129 countries now having adopted various renewable energy policies [5], such as capacity targets [6], tax incentive programs [7], and guaranteed grid connection to renewable electricity sources [8]. However, the growing scale of renewable energy development raises concerns about RET projects exacerbating ongoing environmental and social challenges, potentially leading to conflicts and injustice in host communities. Examples of injustice caused by establishing renewable energy systems include encroachment onto Indigenous people's livelihoods [9] and improper disposal of decommissioned RET components [10]. Such drawbacks

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exemplify the importance of valuing and applying principles of environmental justice in building RETs. An environmental justice framework is useful when identifying and assessing an RET project's impacts, especially on local, poor, and marginalized communities [11]. In our view, promoting environmental justice is essential for ensuring potential environmental and social issues are addressed before RET development and that a policy-oriented framework with restorative justice principles may aid a global effort toward ensuring a just renewable energy transition [12,13].

Multiple studies have been published on the intersectionality of renewable energy and environmental justice. Lacey-Barnacle et al. [14] conducted a review of environmental justice in developing nations, illustrating the roles of decentralization, access, and sustainability for procuring justice. van Bommel and Hoffken [15] applied environmental justice principles to similarly analyze energy justice in contexts within and between European communities. Levenda et al. [16] presented a global scale systematic review of how distributive, procedural, recognition, and capability interpretations of environmental justice are incorporated into renewable energy policy. Studies of environmental justice have also been performed for many individual countries, such as Bulgaria [17], Mexico [18–20], the United States [21,22], Austria [23], Brazil [24], Malawi [25], Morocco [26], and Benin, Senegal, and Togo [27]. Environmental justice studies focused on specific RETs have also been performed for wind [18,28,29], hydroelectricity [30,31], hydrogen [32], biogas [33] and solar energy [26]. However, there exist few examples in the literature of developing a cross-national perspective on RET conflicts linked to environmental injustice, in which past and present injustices guide recommendations for intervention and future prevention. Our research seeks to address this gap in the literature.

Firstly, we examine evidence of environmental injustices in RET development using data from the Global Atlas of Environmental Justice (EJAtlas). Secondly, we produce descriptive statistics and textual summaries based on this evidence to demonstrate how environmental justice has been addressed globally, conveying whether host communities did or did not receive justice or a just outcome in their RET transitions. Finally, we employ the concept of Critical Restoration Geography (CRG) to produce a framework for better understanding documented global occurrences of RET-related environmental (in)justice, and how similar injustices could be avoided for future RET projects. Section 2 provides a background on the global significance of RETs with respect to environmental justice. Section 3 illustrates the methods of data collection, extraction, and derivation of variables from EJAtlas and also explains the concept of CRG. Section 4 summarizes global patterns of RET conflicts, with a specific focus on wind, solar, geothermal, and biomass energy projects and their associated environmental (in)justices. Section 5 examines whether the RET conflicts documented in EJAtlas are true examples of environmental (in)justice, and subsequently how analyzing these examples with our CRG framework can explain why environmental justice was (or was not) served and may be pre-emptively ensured in future conflicts. Section 6 provides conclusive remarks and directions for future development of CRG applications to environmental justice.

2. Background: from global acceptance of renewable energy to environmental justice challenges

Renewable energy is gaining increased attention for its potential to reduce fossil fuel dependency on fossil fuels [34] and to mitigate climate change impacts [35,36]. This attention has led to rapid RET project installation worldwide. Global installed wind energy capacity increased from 198 GW in 2010 to 837 GW by 2021 [37], contributing 16.4 % of electricity utilized in Europe, 12.8 % in China, and 9.2 % in the United States [38]. A report by the Global Wind Energy Council showed that over 341,000 wind turbines had been built globally as of 2022 [39]. Solar energy has seen similarly rapid growth, with the International Energy Agency [40] reporting that global installed solar photovoltaic capacity reached 820 GW in 2021. Geothermal energy, while less widely adopted than wind or solar, has potential to provide a consistent and reliable power source as demonstrated in Iceland and the Philippines [39]. Biomass, particularly that derived from wood and crop waste, provides a significant renewable energy contribution to rural and less industrialized areas. Adoption of biomass as a RET in these areas requires a less complex transition away from infrastructure powered by liquid fuels [41].

Despite renewable energy's potential to make significant contributions to global energy demand, large expanses of land are often required. For example, the United States' target of meeting 20 % of 2030 electricity demand with onshore wind energy would require more than 50,000 km^2 of land use change [42]. High land use requirements mean that RETs have been associated with various negative impacts, such as limited land use for local agriculture and community subsistence [43], biodiversity losses [44], visual impacts arising from changes in landscapes [45], and claims of negative human health impacts caused by noise of operation and shadow flickering [46]. There also exist non-physical and social impacts associated with RETs, such as a lost sense of place, culture, livelihood, and historical norms as communities are forced to reconcile with changes to their surroundings [10,47]. Additionally, property values may decline due to negative perception of new RET projects, though the causal direction and the reality of this relationship are both disputed [48,49]. Although these social impacts are less tangible than impacts like biodiversity loss, their consequences for local RET acceptance [28] also represent environmental injustice to host communities. The exacerbation of such environmental injustice will, according to Fitzgerald [50], require building coalitions between justice activists as the global renewable energy transition accelerates.

Environmental justice discourse became popular in the late 20th Century for identification and assessment of the impacts of proposed or existing infrastructures on poor, marginalized, and racialized communities [51]. The United States Environmental Protection Agency defines environmental justice as "*the fair treatment and meaningful involvement of all people regardless of race, color, national origin, or income with respect to the development, implementation, and enforcement of environmental laws, regulations, and policies*" [52,53]. This definition arguably places more emphasis on distributive environmental justice, which has necessitated recent studies exploring other environmental justice dimensions, such as *procedural* involvement and *restorative* healing of social and environmental damage to identifiable victims [54,55]. Environmental justice also provides a means of understanding and strategizing decolonization efforts [56], with many non-Western and Indigenous perspectives now published that theorize previously disregarded or silenced pathways to environmental justice [57,58]. The prevailing consensus in the literature is that instances of exclusion, unfair treatment, or marginalization of individuals in decision-making processes related to land use, in this case of RET development, may be indicative of environmental injustice. Mueller and Brooks [11] reported three main theoretical explanations for environmentally unjust development of infrastructures or projects: economic, social, and racial. The economic explanation poses that pursuit of profit and return on investment drives industries to develop cheaper land that is often populated by poorer communities [59]. The social explanation is associated with the power of resistance, such that communities lacking resources to resist are more likely to be targets for development [60], a particular problem in countries with authoritarian governments that choose administrative efficiency over justice [61]. Finally, the racial explanation poses that infrastructural development is disproportionately located in places with majority Black, Indigenous, and People of Color (BIPOC) populations. As a result, majority-BIPOC communities are often exposed to higher energy-related health risks, such as pipeline spillages [62] and respiratory illnesses [63], while simultaneously subject to high rates of energy poverty [64].

Ensuring that RETs, or any infrastructural development, do not cause environmental injustice (whether economic, social, or racial) necessitates thinking about how injustices can be avoided in the first place. To

that end, we present Critical Restoration Geography (CRG) as a framework for how previous efforts to restore communities impacted by environmental justice to an equitable state can guide future restoration and prevention efforts. We define CRG here as an applied effort to understand the environmental, social, and political dimensions of restoring communities impacted by environmental injustice. Emerging examples of CRG thinking exist across the literature, particularly those focused on how restoration projects can reproduce or challenge existing power dynamics, such as those of race, gender, and class inequality [65,66]. Doyle and Havlick [67] expanded the scope of CRG beyond physical and environmental features to extend to human elements such as economics, ethics, and politics. Additionally, Anderson and Woelfle-Erskine's [68] study on pedagogies of CRG reported that the intertwined ecological and social dimensions of restoration should go beyond ecosystem repair, extending to the support of human lives and well-being. Avoiding environmental injustice due to RET development would thus require an understanding of how humans interact with the land prior to development, as well as how the restoration of land to a sustainable and/or equitable state may coexist alongside RETs. Smith [69] similarly described restoration as requiring equal placement of value in humans and the environment in their interactions. As such, CRG's emphasis on pre-emptively avoiding future environmental injustice must accommodate ideas beyond simply restoring the land to a prior state (which might not have initially been just), such as new RET developments being coexistent with justice served.

3. Data and methods

3.1. Data source – global atlas of environmental justice (EJAtlas)

We used EJAtlas to collect documented cases of environmental injustice associated with conflicts in RET development from around the world. EJAtlas is an online tool created by the Rosa Luxemburg and Heinrich Böll Foundations in 2011, which allows users to explore and visualize global environmental justice issues, whether of agricultural practice, conservation, industrial activity, or renewable energy development [70]. EJAtlas contains details concerning over 3800 environmental conflicts and cases from more than 120 countries, with evidence for each case collected from news articles, academic papers, non-governmental reports, and other sources [71]. An in-house editorial team confirms the robustness of each source and updates the database frequently. EJAtlas allows users to search for environmental conflicts by location, type of conflict, and industry, while also filtering the data by year, country, and region. One of EJAtlas' key advantages is its ability to highlight patterns of environmental injustice, since EJAtlas can illustrate certain regions or communities that are disproportionately affected by specific conflicts. For instance, EJAtlas' many documented examples of industrial and utility-related injustices in China are consistent with the nation's ongoing challenge to control air and water pollution [72], the same being true of ore mining injustices illustrated across nations in the Andes mountains [73]. This information can be used by academics, activists, and policymakers to support advocacy efforts and policy changes targeted at addressing environmental injustice [53]. It is also worth mentioning that conflicts documented in EJAtlas are generally grounded in objective verification and justified by empirical evidence, rather than being solely reliant on subjective perceptions.

While EJAtlas can highlight patterns and convey overarching reasons for why RETs have historically caused environmental injustice, there are limitations to applying its database for our research purposes. One of the key limitations is EJAtlas' scope, since its database relies on publicly available information that may not capture the full extent and frequency of environmental conflicts and protests. This limitation results in the underrepresentation of certain RETs in the database and limited examples of conflicts in certain countries. Another limitation is that some communities may also be less likely to report or document environmental injustices, which can lead to further underrepresentation in the database. Furthermore, EJAtlas' focus on conflicts and protests means that underlying causes of associated environmental injustice are not directly conveyed, such as structural inequalities, power imbalances, and environmental racism, hence the value of CRG for contextualizing the importance of the database's contents [71]. All of these reasons place limits on the robustness of statistics derived using EJAtlas, as well as the global applicability of CRG to understanding RET conflicts and their associated injustices.

3.2. Data preprocessing - extraction, variables, and analysis

We extracted global examples of environmental (in)justice associated with RETs manually, starting from the year 2001 (EJAtlas' earliest recorded RET conflict) through to 2023. Most of these extracted data were recent conflicts and reflect current global realities of injustices in RET development. Throughout the extraction process, we sought data about conflicts caused by four specific RET types: wind, solar, biomass and geothermal. Given this study's interest specifically in renewable energy resources, which are "replenished at a rate faster than they are consumed" [74], nuclear energy's dependence on finite reserves of fissile metals [75,76] justifies its exclusion. Moreover, despite being an RET, we have excluded hydroelectricity due to existing studies of global hydroelectricity injustices using EJAtlas [77,78]. EJAtlas documents environmental (in)justice pertaining to each of the four chosen RETs using 21 different variables, which we have summarized below in Table 1 under five broader categories (see Supplementary Material for RET conflict details extracted from EJAtlas):

We also computed descriptive statistics to summarize documented RET conflicts and to consequently identify trends of global environmental (in)justice in RET development and siting. Computed statistics include the frequency of conflicts per RET type reported in EJAtlas, presented as annual and cumulative numbers of RET conflicts, along with the country, geographical distribution, the affected population, conflict intensity, project status, and outcome of each conflict. While these statistics serve to identify global trends in when and where RET conflicts have been associated with environmental (in)justice, EJAtlas does not contain every single conflict to have ever occurred across the planet, and some of the available conflicts are not fully documented. However, the documented cases presented here still provide context for how CRG can assist in developing pre-emptive strategies for avoiding future environmental injustice.

3.3. Steps to critical restoration geography (CRG) policies

We developed the CRG framework presented here from collective readings prepared for a symposium by CritRest, a forum of scholars and activists whose intersectional research cuts across racial, disability, and environmental justice movements [79]. We used this reading list, summarized below in Table 2, as a foundation for how the CRG framework was developed and could address issues relating to environmental justice, transcending its prior applications to studies of land degradation and water pollution. Another goal of the reading list is to connect the past and future in the present through restoration geography, since repairing social and environmental damage and striving for justice both require presently making restorative efforts through applying CRG [55]. Our CRG framework thus has important policy implications for resolving RET conflicts and thus ensuring environmental (specifically restorative) justice, including:

- Consideration of the socio-environmental consequences of RET projects, thereby addressing potential conflicts and ensuring sustainable RET siting practices [80].
- Prioritizing equity through fair benefits and engaging with local communities, thus ensuring stable and responsible access to essential materials [81].

Table 1
Variables for each RET conflict documented in EJAtlas, grouped under five categories.

Background	Affected land and demographics	Conflict intensity	Conflict impacts	Conflict outcome and status
Type of RETs	Project area	Level of conflict intensity	Environmental	Status of the project
Name	Affected land area		Social	Status of environmental justice
Country	Affected population		Health	
Specific location	Groups mobilizing for justice			
Year	Specific resources involved			
Project description				
Developers/company				
Type of conflict				
Start date of the conflict				
End date of the conflict				

Table 2
Collective reading list from the CritRest symposium.

Authors	Title	Reference
Christian Keeve, Erin Clancy, Karen Kinslow, and Kallista Bley	Meandering through Critical Restoration Geographies: A Reading Group Collective	[79]
Christopher Reimer	"I would make a home for them": On Crip Interdependence and Abolition Ecology	[83]
Julia Wagner	Restorative Infrastructures	[84]
Robert M. Anderson and Cleo Woelfle-Erskine	Toward Pedagogies of Critical Restoration	[68]
Manon Lefèvre	You'll Find Him in the Sea	[85]
Justyn Huckleberry	Reimagining Mining Operations Toward Local Ownership in the Kalahari Copperbelt	[86]

- Developing legislation that prioritizes the land-energy nexus by exploring the intricate relationships between land use, politics, the environment, and social concerns [82].

Our CRG framework is similar to Sovacool and Dworkin's [87] presentation of a decision-making tool for identifying economic, cultural, environmental, and sociopolitical problems caused by energy use/production. We apply CRG in a similar manner, though with the explicit intent of pre-emptively avoiding future cases of global environmental injustice caused by RETs. Our work achieves this goal by using CRG's principles to explain successes and failures at achieving environmental justice documented in EJAtlas. We subsequently propose how decision-makers and actors engaged in RET development could use these same principles to ensure that future RET projects do not perpetuate similar injustices before, during, or after development. We have derived seven principles that together comprise the CRG framework, and in doing so we have engaged with recommendations for performing good energy social science, such as generalizability of each principle at a global scale and grounding each principle in evidence from the literature [88]. It should also be acknowledged that a framework that attempts to generalize pathways to environmental justice encounters issues of cross-national research, since prevailing causes of RET conflict may be country-specific, or because of differing national progress in making renewable energy transitions [89]. Nonetheless, each principle describes its own means of pre-emptively addressing potential and actual environmental injustices of RET development, thus providing planning and implementation guidance throughout an RET project's life cycle. These principles are:

1. *Recognize the role of power dynamics:* Power dynamics are deeply embedded in RET development, especially in decision-making, access to resources, and in shaping priorities and narratives within RET siting. For example, Heynen et al. [90] emphasized that power imbalances influence not only who participates in projects but also the distribution of costs and other benefits associated with the project. Similarly, Sneddon et al.'s [91] study of strategic relational approaches to state power and civil society highlighted that projects are subject to various power struggles both within and between different stakeholders. As such, understanding the flow of power dynamics by critically examining how power is exercised, negotiated, and contested throughout an RET project's life cycle is important in addressing any form of injustice and ensuring inclusive and effective governance [92,93].
2. *Acknowledge the interconnectedness of social and ecological systems:* RETs have often been depicted as an ecological solution, though this platitude often neglects the social dynamics of RET development [94]. The CRG framework emphasizes that social and ecological components are inseparable and must be considered together for RET development. There must be explicit acknowledgement that human society is deeply intertwined with ecosystems, and that ecosystems do not exist in isolation from human activities [95]. For example, human actions such as land use change and resource extraction both have profound impacts on biodiversity and the behaviour and ecology of animals and plants, with changes to that same ecosystem exerting feedback and on food availability and general human well-being.
3. *Adopt a place-based approach:* In siting RETs, it is important to understand, recognize and respect the unique characteristics, histories, and dynamics of chosen development sites [96]. By adopting a place-based approach, developers and project proponents must recognize that ecological processes, social dynamics, and cultural heritage are distinct and not generalizable. Questions such as "What are the historical, social, and environmental factors that have shaped the community?" and "What are the needs, challenges, and opportunities of the community?" all need to be explored. Considering a community's values, aspirations, and priorities will help ensure that RET development aligns with a community's self-concept of place [97]. Furthermore, a place-based approach to RET development may facilitate a sense of place, connection, and stewardship between communities and all relevant actors. With this approach comes recognition of the relevance of cultural identity, cultural continuity, and a sense of belonging associated with the land [98].
4. *Incorporate multiple knowledge systems:* This principle recognizes that RET development is not limited to scientific knowledge alone. Explicit value must be placed in the contributions and impact of Indigenous and local knowledge, which have a long-standing relationship with the land [99]. Integrating scientific, local, and Indigenous knowledge into RET development enhances the ecological and social well-being of the community through the consequent incorporation and empowerment of diverse perspectives on best practices [100]. Also, engaging multiple knowledge systems means developers can create a platform for dialogue, collaboration, co-learning, and a common shared decision-making forum among different knowledge holders. RET development can thus benefit from a more comprehensive understanding of Indigenous and local knowledge, which often convey ecological insights, species interaction, ecosystem dynamics, and traditional land management that modernist approaches to science cannot [101].

5. *Prioritize equity and social justice:* Most communities have experienced some form of social, economic, or environmental injustice in their past that negatively impacted their well-being and ecosystems [102]. Until these injustices are discussed and addressed, it would be unwise to develop an RET project within these same communities. This principle thus recognizes and understands the historical and current injustices that have impacted specific communities and demographics, which include but are not limited to the legacies of colonization, systemic discrimination, and the unequal distribution of resources and power. RET developers and project proponents must first understand before development begins how communities have been marginalized, as well as the environmental degradation and ecosystem service losses that have consequently occurred [103].
6. *Promote resilience:* As RET development is likely to create disturbances and disruptions, it is important to develop measures that enable communities and ecosystems to adapt, persist, and reconsolidate in the face of these challenges [104]. This principle emphasizes the need for developers to enhance the capacity of socio-ecological systems to withstand and recover from disruptions and disturbances arising from RET development [105].
7. *Modified Empowerment Evaluations (MEE):* Once the above principles have been operationalized for RET development, MEE should be integrated to systematically assess and analyze the outcomes and impacts of RET projects over time [106]. This final principle involves continuous collection and analysis of data and project progress to identify successes and failures, and consequently develop adaptive management strategies, capacity-building, and actor involvement. MEE plays a vital role in RET development because it ensures the effectiveness, efficiency, and long-term sustainability of a given project [107]. By enlisting MEE, developers can easily track challenges and effect corrections with the support of host communities before any severe impact of conflict occurs.

To demonstrate CRG's use in practice, these seven principles will be applied to four case studies of RET conflict extracted from EJAtlas, to better understand how and why environmental justice was (or was not) served in each case. Applying these principles will reveal valuable insights into how environmental injustice transpired from each RET conflict, and thus how informed, context-specific strategies could preemptively avoid future similar injustices while also accommodating equitable, resilient, and sustainable RET development.

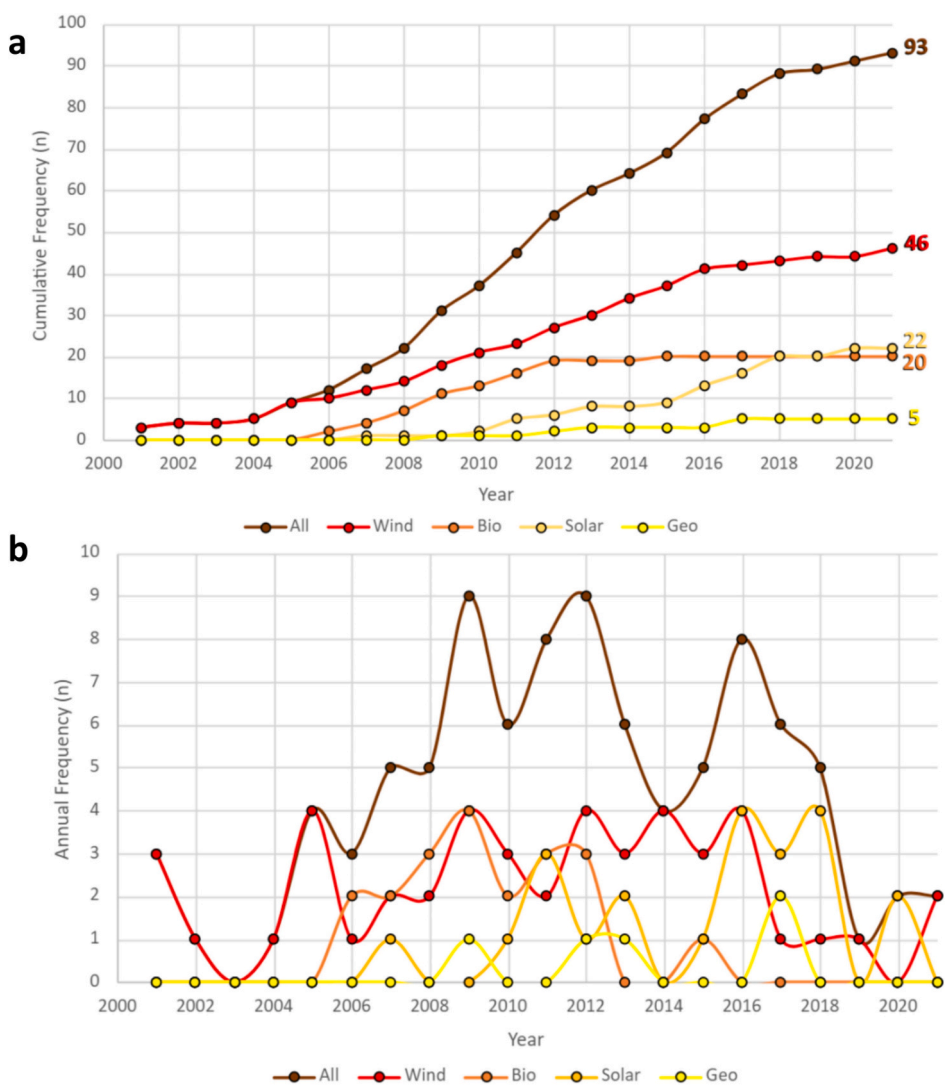

Fig. 1. Cumulative total (a) and annual frequency (b) of reported cases of Renewable Energy Technology conflict in the Global Atlas of Environmental Justice, broken down by all cases (All, brown), wind energy only (Wind, red), biomass energy only (Bio, orange), solar energy only (Solar, light orange), and geothermal energy only (Geo, yellow). NOTE: Nine of the 102 cases of RET conflict lacked a specified year of occurrence (N/A), and thus are not included in this Figure. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Results

4.1. Distribution of environmental injustice cases across the four RETs

Our extraction of documented RET conflicts returned 102 reported cases of environmental (in)justice caused by the four chosen RETs. These reported cases covered the years 2001 to 2021 (zero cases were reported for 2022 and 2023; nine of the 102 cases lacked a year of reported conflict), with the largest number pertaining to wind energy projects (48 out of 102 cases); EJAtlas less commonly reported conflicts about solar (24), biomass (24) and geothermal (6) projects. Fig. 1a and b illustrates the cumulative total and annual frequency of these reported cases of RET conflict. While the cumulative total of reported cases has steadily increased, their annual frequency peaked between 2009 and 2012, and all cases until 2005 were for wind energy conflicts. Moreover, annual cases of biomass and geothermal energy conflicts dropped to zero after 2012 and 2017, respectively, alongside a sustained increase in solar energy conflicts. These inconsistent trends suggest a potential reduction in new biomass and geothermal projects and/or conflicts, or an under-reporting of RET conflicts to EJAtlas, the latter evidenced by the spatial distribution of reported conflicts in Fig. 2. Across the 37 represented countries, Mexico possessed the highest number of reported RET conflicts (16), followed by the United States (10), India (9), France (6), Brazil (5), Kenya (5), Mozambique (3), Sierra Leone (3), and Morocco (3). All other countries with reported cases of RET conflict possessed either one or two examples (see Supplementary Material), likely due to under-reporting in these countries.

4.2. Affected population and conflict intensity

The nature and intensity of the impacts of RET conflicts on communities is shown to vary substantially in EJAtlas' database. Of the 102 relevant conflicts, Table 3 shows that 78 of them (76 %) occurred in rural communities, most of which were wind energy projects (41 out of 78). The remaining conflicts were mostly reported in semi-urban areas (21 %), comprised of near-equal numbers of wind (7 out of 21), solar (7) and biomass (6) projects. An interpretation of the global environmental (in)justices of RET using a CRG perspective would thus be most relevant to rural conflicts. This relevance is important given the number of commercial and residential RET projects that have historically been developed in rural communities [109,110]. EJAtlas' database considers 50 of these 102 cases (49 %) to have possessed low conflict intensity, with decreasing case numbers for medium (36 %) and high (15 %) conflict intensity. Wind energy projects consistently possessed high counts across all three intensity categories, though medium intensity biomass projects (13 out of 36) and high intensity solar projects (7 out of 16) stood out for having higher counts than low and medium intensity projects, respectively. EJAtlas documents conflict intensity categorically as either latent (no visible organizing), low (some local organizing), medium (street protests and mobilization), or high (widespread resistance, violence, and arrests) [111]. As such, two-thirds of documented biomass projects possessing medium to high conflict intensity (16 out of 24) suggests disproportionate socioeconomic impacts compared to wind, solar, and geothermal projects, biomass' unique impacts including diversion from food production and land acquisition for harvesting the biomass itself [112].

Table 3
Number of reported conflicts split by population type and conflict intensity across all four RETs Classification schemes (Urban/Semi-Urban/Rural; Latent/Low/Medium/High) are those used in EJAtlas.

	Renewable Energy Technologies (RETs)				
	Wind	Solar	Biomass	Geothermal	Total
Population type					
Urban	0	3	0	0	3
Semi-Urban	7	7	6	1	21
Rural	41	14	18	5	78
Conflict intensity					
Latent	0	0	0	0	0
Low	28	12	8	2	50
Medium	14	5	13	4	36
High	6	7	3	0	16

4.3. Reported impacts of RET projects on local communities and indigenous groups

The most common impacts of environmental injustice caused by RET conflicts across EJAtlas are those relating to local communities and Indigenous populations. EJAtlas categorizes each conflict's impacts as either environmental, health, or socio-economic, with some impacts being documented more consistently than others, whether due to their prevalence or inconsistent reporting. Table 4 totals the number of occurrences of each impact by RET type among all 102 documented cases,

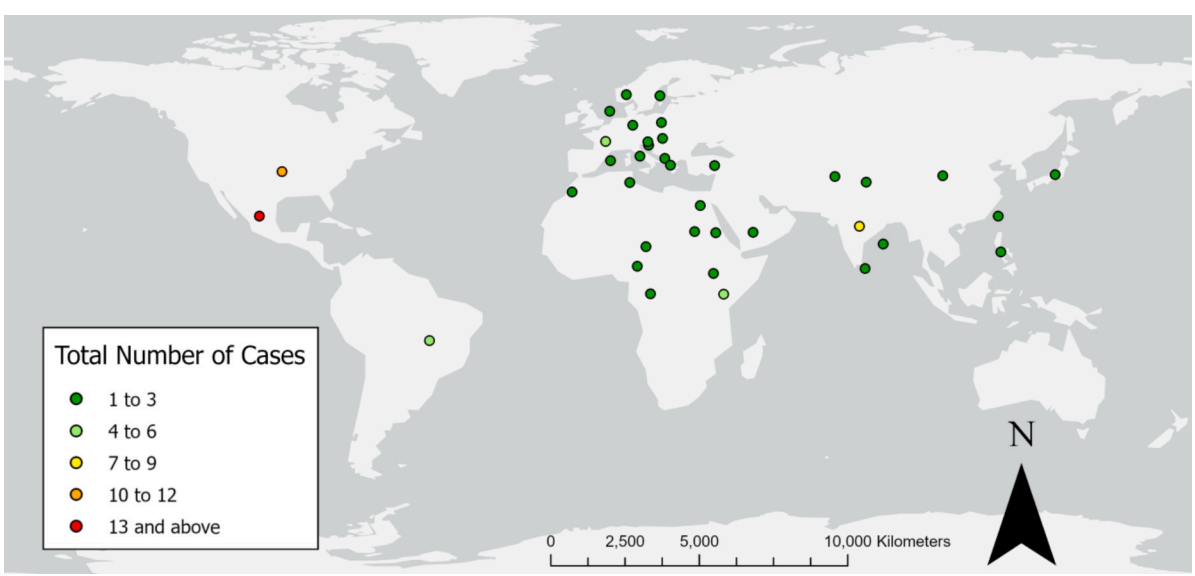

Fig. 2. Global distribution of the 102 reported cases of Renewable Energy Technology conflict in the Global Atlas of Environmental Justice. Basemap from Esri [108].

Table 4
Counts of the frequency of each Environmental, Health, and Socio-Economic impact noted across the 102 cases of RET conflict in EJAtlas. Counts are separated by RET type and given in bold parentheses.

RET type (total conflicts)	Environmental impact (count)	Health impact (count)	Socio-economic impact (count)
Wind **(48)**	Biodiversity Loss **(37)**, Loss or Pollution of Landscape/ Aesthetics **(35)**, Noise Pollution **(30)**, Deforestation and Loss of Vegetation Cover **(18)**, Reduced Ecological/ Hydrological Connectivity **(12)**, Soil Erosion, Contamination, or Pollution **(6)**, Air Pollution **(5)**, Food Insecurity **(5)**, Groundwater Pollution or Depletion **(3)**, Large Scale Disturbance **(2)**, Other Environmental Impacts **(2)**, Surface Water Pollution **(2)**, Damage to Coastal Areas **(1)**, Desertification and Drought **(1)**, Floods **(1)**	Stress **(10)**, Depression **(9)**, Exposure to Radiation or Other Uncertain Risks **(9)**, Accidents **(4)**, Violence Related Health Impacts **(3)**, Deaths **(2)**, Suicide **(2)**, Occupational Disease **(1)**	Loss of Landscape/ Sense of Place **(27)**, Loss of Livelihood **(25)**, Loss of Traditional Knowledge, Practices, and Cultures **(24)**, Displacement **(20)**, Land Dispossession **(19)**, Violation of Human Rights **(18)**, Increase in Corruption of Different Actors **(17)**, Lack of Work Security, Labor Absenteeism, Firings, and Unemployment **(3)**, Militarization and Increased Police Presence **(2)**, Specific Impacts on Women **(2)**, Alcoholism and Prostitution **(1)**, Increase in Violence and Crime **(1)**, Other Socio-Economic Impacts **(1)**
Biomass **(24)**	Biodiversity Loss **(23)**, Disturbance of Hydro and Geological Systems **(23)**, Food Insecurity **(23)**, Loss of Aesthetic **(22)**, Soil Contamination **(22)**, Large Scale Reduced Ecological/ Hydrological Activity **(22)**, Groundwater Pollution **(1)**, Noise Pollution **(1)**	Malnutrition **(9)**, Deaths and Other Environmental Related Diseases **(7)**, Occupational Disease and Accidents **(5)**, Rape, Homicide, and Other Violence Related Health Impacts **(4)**, Exposure to Radiation or Other Complex Health Risks **(3)**, Other Health Impacts **(3)**, Health Problems Related to Alcoholism, Prostitution **(2)**, Mental Problems Including Stress, Depression, and Suicide **(2)**	Loss of Landscape and Sense of Place **(23)**, Displacement **(21)**, Loss of Livelihood, Traditional Knowledge, and Cultures **(21)**, Violation of Human Rights **(21)**, Neglect of Local Workers **(1)**, Socio-Economic Inequalities in Hiring, Leadership, and Power Control **(1)**
Solar **(24)**	Biodiversity Loss **(23)**, Loss of Aesthetic **(18)**, Soil Contamination **(18)**, Food Insecurity **(18)**	Exposure to Radiation **(7)**, Accidents and Other Environmental Related Diseases **(6)**, Malnutrition **(2)**, Stress, Depression, and Other Mental Health Problems **(2)**,	Loss of Landscape/ Sense of Place **(21)**, Displacement **(18)**, Loss of Livelihood **(18)**, Loss of Traditional Knowledge and Cultures **(18)**, Violation of Human Rights **(17)**
Geothermal **(6)**	Disturbance of Hydro and Geological Systems **(6)**, Biodiversity Loss **(5)**, Food Insecurity **(5)**, Large Scale Reduced Ecological/ Hydrological Connectivity **(5)**, Loss of Aesthetic **(5)**, Soil Contamination **(5)**, Induced Seismic Activity **(1)**	Accidents **(1)**, Deaths **(1)**, Pressure Against Oppositional Forces **(1)**, Reduced Community Trust **(1)**, Stress due to Risk of Seismic Activity **(1)**	Loss of Landscape/ Sense of Place **(5)**, Loss of Traditional Knowledge/ Practice/Cultures **(5)**, Displacement **(3)**, Loss of Livelihood **(3)**, Violation of Human Rights **(3)**, Burden of Proof of Damages Place on Plaintiffs **(1)**

revealing that certain impacts are documented frequently across all four types. These common impacts include biodiversity loss, various forms of pollution (noise, aesthetic, landscape), claims of depression and elevated stress, loss of sense of place and traditional knowledge, and displacement from native lands.

Table 4 also shows some documented impacts to be unique to specific RETs, such as geothermal's disturbance to hydro/geological systems and biomass' associated spread of malnutrition. Nevertheless, these unique impacts are in the minority, suggesting a consensus of what environmentally unjust RET development broadly looks like. It should be noted that the impacts documented in EJAtlas (besides being under-reported) do not always use consistent language (e.g., soil erosion versus soil degradation), impacts may be combined as a single impact (e.g., displacement versus displacement and loss of livelihood), and their inherent subjectivity means that perceived injustices may not reflect scientific understanding (e.g., perception of noise pollution from wind turbines is often greater among people with preconceived negative opinions [113]). These findings place limits on how generalizable CRG's application is for understanding case studies of environmental (in) justice.

5. Discussion

5.1. Interpreting EJAtlas and the attainment of environmental justice

Global interest in renewable energy development is largely driven by pressures to meet domestic energy demand and to address the threat of climate change, though EJAtlas reveals that these pressures often lead to inadvertent negative impacts and environmental injustice in host communities. EJAtlas' database specifically details the geographical conditions, consequences, and in some cases potential solutions to various forms of global environmental injustice, with injustices over RET conflict being the focus of this work. Locations for common RET conflicts are Mexico and the United States, for which many examples exist in EJAtlas for wind and solar energy conflicts (Fig. 3), Mexico is characterized by uneven geographies of energy consumption, thought to contribute to a corresponding unevenness in RET conflicts. For example, affected communities have explicitly contested wind energy development in Mexico's windiest regions due to preference to supply energy to several multinational corporations, rather than these local communities [19]. Moreover, the largest numbers of biomass energy conflicts have occurred in Kenya, Mozambique, and Sierra Leone (Fig. 3), countries in which extensive palm oil and sugarcane plantations are increasingly being used for biomass energy production, posing a threat to the rich biodiversity in these countries and thus a heightened risk of environmental injustice [114]. Fig. 1b suggests that the annual frequency of global RET conflict neither increased nor decreased overall since 2001. This lack of trend could be attributed to under-reported cases, particularly for biomass and geothermal projects, but also to conflict reduction

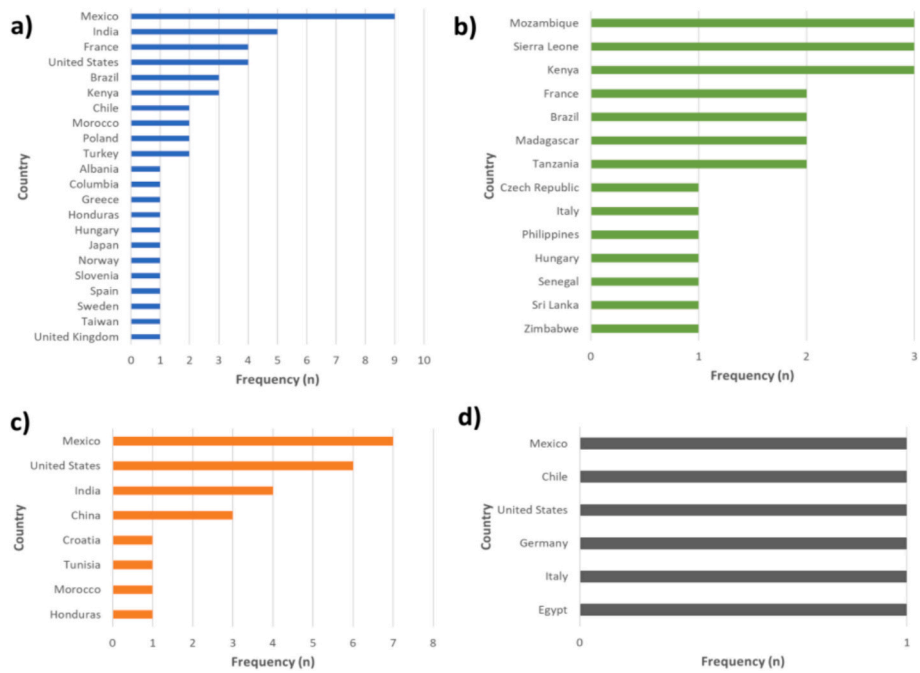

Fig. 3. Number of reported cases by country of Renewable Energy Technology conflicts in the Global Atlas of Environmental Justice database. From top-left to bottom-right are the reported cases for wind energy (a), biomass energy (b), solar energy (c), and geothermal energy (d).

in some countries being negated by negligent RET development in others, limiting the statistical interpretation of EJAtlas' database. However, the database suggests that most reported RET conflicts result from forceful land acquisition by governments and private sector corporations, which has often led to host communities losing their land rights. Indeed, some of the most frequently reported impacts due to this acquisition (Table 4) include loss of biodiversity and landscape (environmental), exposure to radiation and visual impacts (health) and loss of sense of place, culture, and attachment (social). These findings are consistent with many reported and published examples of environmental injustice literature [1,13,33,114–118].

Despite the many documented cases of environmental injustice in EJAtlas, multiple examples exist in the database of justice having been served in some affected communities. Fig. 4 summarizes the number of RET conflicts that ended with justice served and whether the strategies to achieve justice resulted in the RET project being canceled or allowed to continue. Twenty-seven (26.5 %) of the 102 RET conflicts ended with environmental justice served, while not served in 55 (53.9 %) conflicts and being ambiguous in the remaining 20 (19.6 %). Of the 27 conflicts that ended with justice served, 20 (75.2 %) of them resulted in the RET facility being closed, four facilities (16.7 %) are still in operation, and the final three (11.1 %) still have plans to progress.

EJAtlas provides common justifications for whether justice was served across these RET conflicts, as summarized in Table 5. Using the same categories for justice served in Fig. 4 (Yes/No/Not Sure), the achievement of environmental justice was most often based on closure of RET facilities, RET project suspension, revision of legislation, or improved consultation between all key actors involved. RET conflicts with ambiguous (Not Sure) justice were typically relocated, had ongoing or unfinished environmental impact assessments, featured partial or inadequate local participation in project modification and resistance, or made only some land restoration effort. EJAtlas considered justice to have not been served in cases with ongoing RET conflicts, such as the use of brute force, oppression of communities and their demands, governmental control, or persistent ecological impacts. Given the non-trivial number of justice outcomes classified by EJAtlas as "Not Sure" (Fig. 4), we suggest the use of multiple perspectives of what environmental justice can look like for RET conflicts to reduce this ambiguity.

One option for diagnosing ambiguous justice is to enlist Carrillo and Pellow's [119] "pillars" of environmental justice, which account for

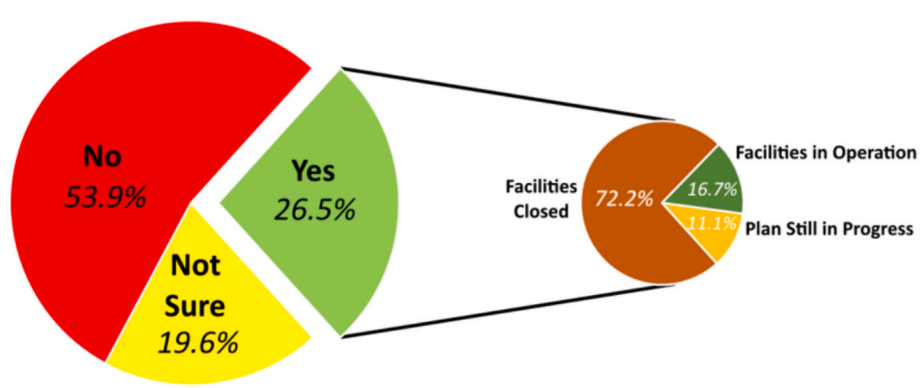

Fig. 4. Percentage of different environmental justice outcomes for the 102 Renewable Energy Technologies conflicts in the Global Atlas of Environmental Justice. The conflicts for which environmental justice was served (Yes) are further separated by how RET project development was consequently impacted (Closed, In Operation, Plans Still in Progress).

Table 5
Justifications given by EJAtlas for examples of environmental justice (not) being served across all 102 documented RET conflicts.

Was environmental justice served?	Common justifications from EJAtlas
Yes	Project cancellations, suspensions, and demolition of RET structures Revision of legislation for public participation in decision-making Engineering controls including construction of structures to protect biodiversity, such as birds. Improved alliances between government, community members, local and foreign conservationists
Not sure	Relocation of project from original sites to new areas with less resistance. Project halted but the companies still own the land, so there is a probability of future projects. Project ongoing with improved environmental impact studies Decision on hold, case still in court Attempted reforestation of large hectares of land to make up for the large land mass deforested to construct the RET Partial involvement of local opinions.
No	Projects are ongoing despite protests, conflicts, and resistance. Politicization of media leading to the refusal of local media to cover news. Criminalization of protests leading to feelings of powerlessness Government backing includes the exclusion of local opinions in decision-making and the provision of subsidies to RET companies. Government is protecting business interests at the expense of the human security needs of the local people. Ongoing displacement, land grab, occupation, and resource exploitation. Use of the Police Force to displace people with violence and coercion to re-acquire the land. Continuous human and ecological impacts including noise pollution, decreasing property values, shadow flicker, and wildlife impacts.

interdependent forces in achieving justice to thus critique purported efforts and outcomes. Each of these four pillars is summarized below, with examples given of each one in the context of RET conflict:

1) "Intersecting social categories, including race, class, gender, sexuality, and species." This pillar accounts for the human domination of nonhuman species that results in ecocide and natural habitat destruction. Example: the land transformation required to build wind farms and valuing the consequent impacts on both avian species and human discomfort.
2) "Multi-scalarity regarding space and temporality", thereby recognizing that environmental injustice can happen over different timeframes and spatial domains, separately and simultaneously. Example: recognizing that biomass production's impacts range from local biodiversity disturbance to potentially global consequences for atmospheric pollution.
3) "State power that reproduces the social inequalities inherent to capitalism," and the responsibility of state and national governments to recognize their role as actors that have historically and structurally enabled these inequalities. Example: ending cycles of environmental racism by deliberately not siting RET projects or their waste products in the vicinity of majority-BIPOC populations [120].
4) "Perceived expendability of marginalized populations," instead recognizing the value, indispensability, and traditional knowledge that these populations (and nonhuman species) possess. Example: actively consulting with and prioritizing the land rights of Indigenous people before, during, and after the RET siting process [121,122].

These four pillars complement the framing of environmental justice as also having four dimensions: *procedural* justice as the inclusion of communities in decision-making processes on issues that concern them; *distributive* justice as the allocation of rights, duties, advantages, and disadvantages of the actions of institutions; justice as the *recognition* of sociocultural, ethnic, racial and gender differences; and the *restorative* justice of reasserting the health, safety, and dignity of victims of injustice [123,124]. We can use these four dimensions/pillars of environmental justice to evaluate the ambiguity of whether justice was served following RET conflict based on the reasons for ambiguity summarized from EJAtlas in Table 5. For instance, while relocation of a geothermal plant may achieve procedural justice if relocation was motivated by a host community's resistance, this relocation would lack distributive justice if the generated energy becomes inaccessible to said community. As another example, a private company replanting trees following a wind farm's cancelation recognizes the non-expendability of ecosystems and is therefore restoratively just, but conversely lacks recognition justice due to exclusion of local land rights holders. These multiple dimensions of assessing environmental justice, and the existence of common means of serving justice across different countries, is precisely what facilitates our CRG framework's potential to inform policy for how to pre-emptively ensure environmental justice is served for RET development.

5.2. *Using critical restoration geography to prevent future environmental injustice*

The conclusion that environmental justice has been served in an area and for a people requires assessment of all dimensions of harm caused by current and potential RET facilities, followed by development of strategies that continue to implement environmental justice moving forward. Therefore, we propose the use of CRG as a guiding framework for how to understand causes of historical RET conflicts and to consequently inform policies that would avert conflicts for proposed, existing, and future RET projects, thus serving environmental justice. The concept of CRG has previously been applied mostly to assessing restoration programs, development projects, and post-disaster landscapes [125–128]. We posit here that CRG can also be applied to any form of pre-infrastructural development, such as that of RETs, as progress toward an equitable system would integrate environmental, social, economic, and political considerations to enable a thriving society. We present here four case studies of RET conflict derived from EJAtlas, each serving as a different example of environmental justice (not) being served, and subsequently showing how CRG's principles can explain the outcomes of each conflict.

5.2.1. *Case studies for which environmental justice led to RET project cancelation/suspension*

Although Table 5 posits both RET project cancelation and suspension as means for serving environmental justice, we argue that these outcomes alone do not suffice, given recognition injustices persisting long afterward in some instances. The Dakatcha Woodland Biofuel Project in Kenya serves as a poignant example [129]. As one of Kenya's largest biofuel initiatives, preparation to grow jatropha crops in this area began in 2009 without undergoing a comprehensive Environmental Impact Assessment (EIA). The Dakatcha Woodlands also represent a biodiversity hotspot and home to over 20,000 people from the Watha and Giriama tribes. These local tribes, deeply connected to the woodlands, raised objections and pursued legal action to halt land clearance and construction for the project. Despite a court ruling to cease operations, untreated industrial wastewater continued to be discharged into the community river, endangering the local ecosystem and inhabitants. Applying our CRG framework's principles prior to commencing land clearance, such as the role of power dynamics in RET projects (Principle 1) and promoting resilience (Principle 6), could have averted this recognition injustice in the host communities. Firstly, since the power dynamics of RET development are embedded in decision-making, access

to resources, shaping priorities and narratives, and finally adequate consultation and resource extraction mechanisms should have been discussed before land clearance began. Secondly, promoting resilience was important since the clearance of the Dakatcha Woodlands would have certainly created ecological disturbances and disruptions. Developing measures of resilience would have enabled the local tribes and ecosystems to adapt, persist, and reconsolidate in the face of challenges created by industrial wastewater disposal.

There are, however, examples of cancelation or suspension that unambiguously led to environmental justice served, such was the case for the proposed development of wind turbines close to Mont-Saint-Michel in France [130]. An iconic UNESCO World Heritage Site renowned for its natural beauty, Mont-Saint-Michel stands as a testament to centuries of cultural and natural evolution, unique characteristics, historical significance, and delicate ecological balance, making it a focal point for conservation efforts. While all necessary permits for the wind turbine construction were granted by a prefectural decree, various groups of actors, including the Environment and Landscape Association in Upper Brittany and Ille-et-Vilaine, as well as owners of the nearby Château de la Ballue, contested this decision in administrative court. UNESCO also expressed concern, threatening to remove Mont Saint-Michel from its historical monument list due to a wind farm's potential landscape distortion and adverse effects on local ecology and ecotourism. The courts ultimately ruled against the project in 2013, hence procedural justice was served to both conservation activists and Mont-Saint-Michel's inhabitants. Applying the CRG principle of adopting a place-based approach (Principle 3) could have, however, mitigated the concerns, conflict, and injustice surrounding the proposed wind turbines. A project plan rooted in an understanding of Mont Saint-Michel's distinctiveness would have guided its developers toward recognizing the intricate interplay of ecological processes, social dynamics, and cultural heritage. Such recognition may have led to fewer, smaller wind turbines being proposed, or the project being set further back from Mont-Saint-Michel, reducing its risk of being canceled by the courts. Ultimately, CRG acknowledges the non-generalizability of factors pertaining to RET development, allowing proponents and planners to serve communities in environmentally just ways that safeguard local, historical, and environmental integrity.

5.2.2. Case studies for which environmental justice led to RET project rehabilitation/restoration

EJAtlas also evidences that justice served may allow RET projects to coexist without cancelation, given necessary measures and plans to rehabilitate, and in some cases restore, a landscape to its original state. One example is the Ivanpah Solar Power Electric Generating System in California [131], which at the time of construction (2012) was the largest concentrated solar power plant in the United States. Occupying approximately 3500 acres of the Mojave Desert, the project was intended to double the United States' total solar electricity output. However, environmental activists and Indigenous communities expressed concerns about Ivanpah's impact on wildlife, specifically on the desert tortoise and local bird populations. There were also concerns about engagement with land rights holders and other local actors during project consultations, as not all were consulted before construction began. Due to these concerns, Ivanpah's developers temporarily halted the project and took several measures to ensure that the project valued environmental justice, such as building fences to deter animals and wildlife, decreasing the project's installed capacity by 12 % to protect endangered plants, holding several community participation meetings, and acquiring new EIA permits. These efforts are encapsulated by the CRG framework's principles of incorporating multiple knowledge systems (Principle 4) and acknowledging the interconnectedness of social and ecological systems (Principle 2). Incorporating knowledge possessed by Indigenous communities and local actors is crucial for minimizing an RET project's negative impacts throughout its life cycle, from initial land clearance practices to disposal of damaged or worn components. Also, the decision to scale back the Ivanpah project and install structures to limit ecosystem interference represents some effort at acknowledging the interconnectedness of a social demand for energy at an ecological expense. While not a perfect service of restorative justice, the Ivanpah project represents an attempt at pre-emptively avoiding its worst possible ramifications, as conveyed by the CRG framework.

While the previous example is one of land rehabilitation prior to RET development, sometimes this process occurs later, leading to high conflict intensity and even the demolition of completed projects. This exact situation occurred after the announcement of InfraVest's proposed construction of 14 offshore wind turbines 60–200 m off the coast of Yuanli, Taiwan in 2012 [132]. This announced development sparked opposition from locals and Indigenous groups, which led to over 60 % of Yuanli's residents signing a petition against the proposed wind farm's development, or any future turbine constructions in the vicinity. Following two years of protests, organized blockades, and deliberations, an agreement between InfraVest and the community led to the turbines nearer residential zones being dismantled by 2016, downsizing the project's original power capacity by 30 %. Throughout this period, InfraVest orchestrated various community gatherings and outreach initiatives aimed at fostering awareness and education regarding the project's intricacies. From a CRG perspective, InfraVest eventually did address the power dynamics of wind farm siting (Principle 1) and advocated for a place-based approach (Principle 3). Specifically, InfraVest's engagement of land rights holders and local actors fostered avenues for public discourse and decision-making that addressed its power dynamic with Yuanli's residents, and the removal of select turbines to rehabilitate the landscape met place-based community requisites for the level of wind energy development desired. The procedural and restorative injustice that Yuanli endured could have been avoided had CRG's principles been applied during the pre-siting stage, as they were for the Ivanpah and Mont-Saint-Michel projects. These case studies exemplify the value of CRG for pre-empting instances of inequity and discord, and for developing policy that prevents similar injustices in different contexts. Moreover, these case studies demonstrate that cancelation or suspension of an RET project is not a ubiquitous solution for achieving environmental justice. Rather, the application of CRG's principles for understanding why environmental injustice happens provides a framework for how new RET projects can avoid foreseeable negative impacts.

5.2.3. Comparing CRG against other frameworks of environmental justice (UNSDGs, NEPA)

The seven CRG principles collectively serve as guidelines for how decision-makers involved in RET development can pre-emptively avoid causing environmental injustice. Principles like resilience, social justice, and interconnectedness are general and thus applicable in any infrastructural development context in any location (given the limitations of EJAtlas' dataset, see Section 4.1), hence, we recommend their consideration when planning future RET projects. It is important, however, to clarify how CRG differs from frameworks of environmental justice. Malin and Ryder [133] developed three principles of their own that capture "intersectional environmental injustice". These principles are; 1) Emphasizing multiple social locations and intragroup differences 2). Exploring issues with a multi-scalar lens and 3). Critically identifying and analyzing not only powerful actors but systems and processes of power in these dynamics, the latter also being a key component of Pellow's [134] framework for "critical environmental justice". While there are common features between these principles and those of CRG, such as recognizing the role of power dynamics and incorporating multiple knowledge systems, the CRG framework is broader given its explicit consideration of resilience, prioritizing equity, and adoption of a place-based approach. Another key difference is how CRG and these other environmental justice frameworks mobilize principles of social justice. Whereas CRG focuses specifically on integrating socio-ecological restoration into justice considerations, environmental justice tends to

instead advocate for systemic change in response to unequal treatment [135]. Given CRG's use of seven general principles to construct place-based approaches to justice served, and CRG's conveyance of socio-ecological understanding of past RET conflicts (Section 5.2.2), CRG is perhaps more readily applicable than environmental justice frameworks for guiding policy development that circumvents future conflicts.

The CRG framework also has common and distinguishing features with goal-based and legislative approaches to environmental justice, such as the United Nations Sustainable Development Goals (UNSDGs), and the United States National Environmental Policy Act (NEPA). Some of their commonalities include: 1) An emphasis on solving environmental and social challenges and promoting sustainable development, 2) Advocating for a systematic interdisciplinary approach and interconnectedness of various principles and goals, 3) Acknowledging the importance of inclusivity, involving diverse actors, and community engagement in achieving goals, and 4) Promoting ethical practices by ensuring a just energy transition [79]. Moreover, the CRG framework's application aligns with several of the UNSDGs, especially "Climate Action" (Goal 13) and "Life on Land" (Goal 15) [136]. Despite these commonalities, the CRG framework also differs from UNSDGs and NEPA in important ways. For example, CRG's principles are intended for examination of social dynamics, power structures, and historical injustices of landscapes, in this context for RET development, providing a more nuanced understanding of community needs [137]. Additionally, CRG prioritizes environmental justice by addressing inequities in resource access, decision-making, and RET benefits, while incorporating local knowledge systems into RET planning and implementation [138]. By contrast, the UNSDGs are objective-oriented and explore sustainable development more holistically than environmental justice alone, and NEPA (and other national-level environmental policies) explicitly legislates and sanctions projects that cause environmental injustice [139], which the CRG framework does not. Therefore, while CRG shares common environmental and social objectives with UNSDGs and NEPA, the former is different for being a tool to aid policymaking (unlike NEPA) and for focus on pre-emptive avoidance specifically of environmental injustice (unlike UNSDGs).

6. Conclusion

This paper utilizes the Global Atlas of Environmental Justice to summarize the global state of environmental injustice relating to the development of Renewable Energy Technologies (wind, solar, biomass, and geothermal). Importantly, our work's overall contributions are to summarize documented evidence for global environmental injustice resulting from RET conflicts, and to explain how pre-emptive strategies and policy recommendations pursuant to avoiding future injustice may be developed. Although EJAtlas' database conveys overarching reasons for why RETs have historically caused environmental injustice, it does not explain how future injustices can be avoided, a knowledge gap filled by this paper through its interpretation and application of Critical Restoration Geography. CRG's existing applications focus on understanding why efforts to undo injustices relating to land use and resource extraction have succeeded or failed, though CRG has not been applied either to RET siting specifically, nor to efforts intended to pre-emptively avoid injustice in the development phases of RET infrastructure. Using four case studies from EJAtlas that represent examples of environmental justice (not) being served in different contexts, we have demonstrated how CRG's principles, such as valuation of place-based approaches, acknowledging the role of power dynamics, and incorporating multiple knowledge systems, can explain the outcome of RET conflicts. These principles are global in context and thus universally applicable for renewable energy developers to enlist when conceiving new projects that are cognizant of potential environmental consequences.

The seven CRG principles offer critical implications for RET policy design in conflicted host communities. For example, acknowledging the power dynamics of decision-making is foundational for ensuring accountability and compliance monitoring in an RET project's planning phase, thus avoiding the creation and aggravation of new and existing environmental injustices, respectively. CRG thus enables policymakers to consider the interconnectedness of social and ecological systems, leading to opportunities for both nature protection and nature-based solutions in RET infrastructure development. Policymakers can also use these CRG principles to establish recognition and procedural justice in an RET context, while also enhancing the capacity of socio-ecological systems to withstand and recover from disturbances arising from RET development. We therefore recommend the use of CRG's seven principles to guide policies from local to transnational scale that do not repeat the injustices caused by previous RET development projects, and that future work should focus on the feasibility of CRG's political integration. This work ultimately demonstrates the importance of valuing environmental justice when siting and developing RETs. Therefore, RET developers and researchers should consider environmental justice as a key factor in renewable energy planning and siting, such as in the performance of multi-criteria decision analysis for RET site selection and development.

The authors of this work hereby declare that this work represents original research and hasn't been submitted to any journal for publication. The authors also declare that they all qualify as authors of the submitted work and grant its approval.

CRediT authorship contribution statement

Chinedu C. Nsude: Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Rebecca Loraamm:** Writing – review & editing, Supervision, Investigation. **Joshua J. Wimhurst:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **God'sgift N. Chukwuonye:** Writing – review & editing, Writing – original draft, Formal analysis. **Ramit Debnath:** Writing – review & editing, Resources, Investigation.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

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Corrigendum

Corrigendum to "Renewables but unjust? Critical restoration geography as a framework for addressing global renewable energy injustice" [Energy Res. Soc. Sci. 114 (2024) 103609]

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The authors regret an error in Section 3.1 regarding the origin and funding sources of the Global Atlas of Environmental Justice (EJAtlas) where it says "EJAtlas is an online tool created by the Rosa Luxemburg and Heinrich Boll Foundations" The correct sentence is "EJAtlas is an online tool created and directed by Prof Joan Martínez Alier and Dr. Leah Temper and coordinated by Dr. Daniella del Bene. The EJAtlas was initially supported by the European Commission through the Environmental Justice Organizations, Liabilities and Trade (EJOLT) project. EJAtlas has continued to receive support from volunteers worldwide and various research projects and collaborators, particularly the International Academic and Activist co-production of knowledge for Environmental Justice (ACKnowl-EJ) research project and the ENVJUSTICE European Research Council (ERC) award".

The authors would also like to acknowledge and cite the EJAtlas using the following articles.

1. Temper, L., Demaria, F., Scheidel, A. et al. The Global Environmental Justice Atlas (EJAtlas): ecological distribution conflicts as forces for sustainability. Sustain Sci 13, 573–584 (2018). https://doi.org/10.1007/s11625-018-0563-4
2. Leah Temper, Daniela del Bene and Joan Martinez-Alier. 2015. Mapping the frontiers and front lines of global environmental justice: the EJAtlas. Journal of Political Ecology 22: 255–278. https://journals.librarypublishing.arizona.edu/jpe/article/id/1932/
3. Temper, Leah, Sofia Avila, Daniela Del Bene, Jennifer Gobby, Nicolas Kosoy, Philippe Le Billon, Joan Martinez-Alier, et al. 2020. "Movements Shaping Climate Futures: A Systematic Mapping of Protests against Fossil Fuel and Low-Carbon Energy Projects." *Environmental Research Letters*. IOP Publishing Ltd. http://doi:10.1088/1748-9326/abc197.
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